4DVarNet-FM: Training Experiments (14 runs, E*/F*/G*)

Experiment Hyperparameters

ID	Model	Channels	S1Ep	S2Ep	LR	GradClip	TimeEmb	N_outer	SigPrior	TrainMix	Rand	ParamNoise
E1_direct_unet_default_cs1	DirectUNet	[64, 128, 256]	300	0	0.001	10.0	-	-	-	cs1	no	-
E1_direct_unet_default_cs2	DirectUNet	[64, 128, 256]	300	0	0.001	10.0	-	-	-	cs2	no	-
E2_direct_unet_small_cs1	DirectUNet	[32, 64, 128]	300	0	0.001	10.0	-	-	-	cs1	no	-
E2_direct_unet_small_cs2	DirectUNet	[32, 64, 128]	300	0	0.001	10.0	-	-	-	cs2	no	-
F1_vanilla_cfm_default_cs1	VanillaCFM	[64, 128, 256]	500	0	0.001	10.0	64	10	0.5	cs1	no	-
F1_vanilla_cfm_default_cs2	VanillaCFM	[64, 128, 256]	500	0	0.001	10.0	64	10	0.5	cs2	no	-
F2_vanilla_cfm_small_cs1	VanillaCFM	[32, 64, 128]	500	0	0.001	10.0	64	10	0.5	cs1	no	-
F2_vanilla_cfm_small_cs2	VanillaCFM	[32, 64, 128]	500	0	0.001	10.0	64	10	0.5	cs2	no	-
G1_tweedie_default_cs1	Tweedie	[64, 128, 256]	300	200	0.001	10.0	-	-	-	cs1	no	-
G1_tweedie_default_cs2	Tweedie	[64, 128, 256]	300	200	0.001	10.0	-	-	-	cs2	no	-
G2_tweedie_small_cs1	Tweedie	[32, 64, 128]	300	200	0.001	10.0	-	-	-	cs1	no	-
G2_tweedie_small_cs1_k1	Tweedie	[32, 64, 128]	300	200	0.001	10.0	-	-	-	cs1	no	-
G2_tweedie_small_cs2	Tweedie	[32, 64, 128]	300	200	0.001	10.0	-	-	-	cs2	no	-
G2_tweedie_small_cs2_k1	Tweedie	[32, 64, 128]	300	200	0.001	10.0	-	-	-	cs2	no	-

Shared settings (config/lorenz63_default.yaml):
  dt=0.01, T_max=3.0s, obs_interval=20, R_var=0.5, B_var=2.0
  num_windows=2000, spinup_steps=10000, batch_size=32, dropout=0.1
  optimizer=Adam, gradient_weight=0.1 (grad-loss term)
  Stage 2 only ever runs for Tweedie (S2Ep>0 above); DirectUNet/VanillaCFM are stage-1-only.

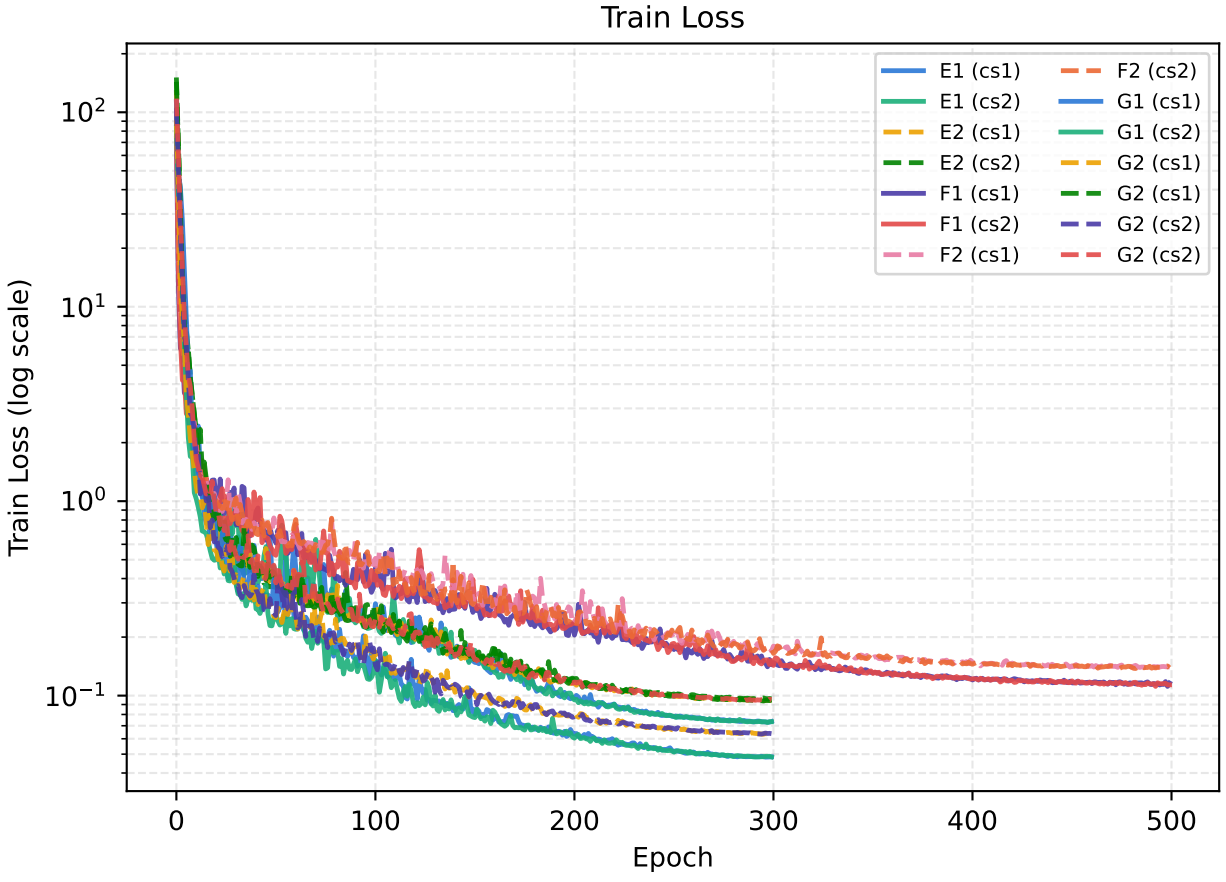
CS1: param_bias=0.0, forcing_state_bias=0.0, coupling=linear (ideal conditions)
CS2: param_bias=0.15, forcing_state_bias=0.15, coupling=quartic (corrupted)
See TrainMix column above for what each experiment was actually trained on.

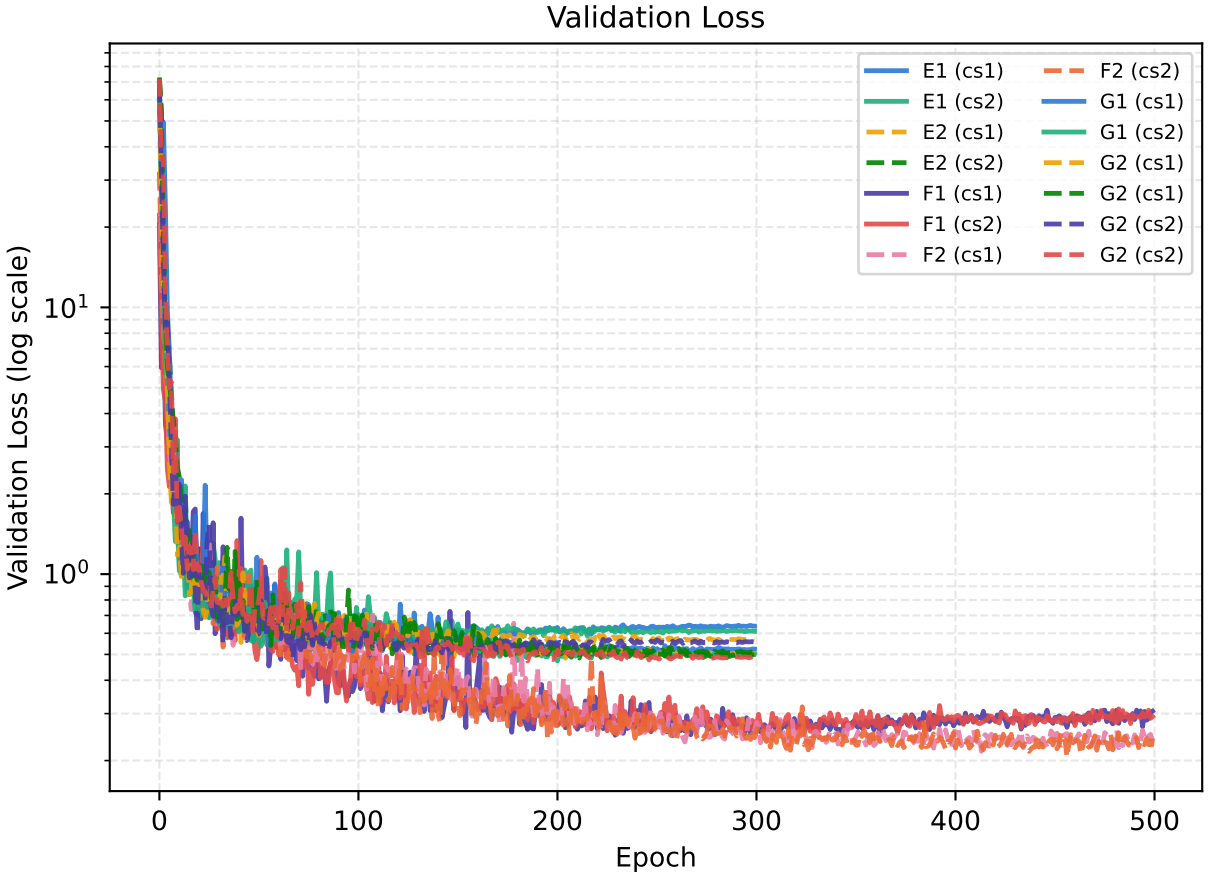
DirectUNet: single UNet1D pass, obs -> state (MSE + gradient loss), no flow matching.
VanillaCFM: conditional flow matching, LinearInterpolant(nu=1.0), Euler sampling
            over N_outer=10 steps, x0 ~ N(0, sigma_prior).
Tweedie: 2-stage solver -- stage1 fits a Gaussian mean estimator, stage2 fits a
         non-Gaussian residual correction on top of the frozen stage1 estimator.

Run Status

E1_direct_unet_default_cs1	trained (results + loss log)
E1_direct_unet_default_cs2	trained (results + loss log)
E2_direct_unet_small_cs1	trained (results + loss log)
E2_direct_unet_small_cs2	trained (results + loss log)
F1_vanilla_cfm_default_cs1	trained (results + loss log)
F1_vanilla_cfm_default_cs2	trained (results + loss log)
F2_vanilla_cfm_small_cs1	trained (results + loss log)
F2_vanilla_cfm_small_cs2	trained (results + loss log)
G1_tweedie_default_cs1	loss log only
G1_tweedie_default_cs2	loss log only
G2_tweedie_small_cs1	trained (results + loss log)
G2_tweedie_small_cs1_k1	trained (results + loss log)
G2_tweedie_small_cs2	trained (results + loss log)
G2_tweedie_small_cs2_k1	trained (results + loss log)

Training & Validation Loss





Results Summary (CS1 / CS2)

Per-Variable Mean RMSE Summary (CS1 / CS2)

--- CS1 ---

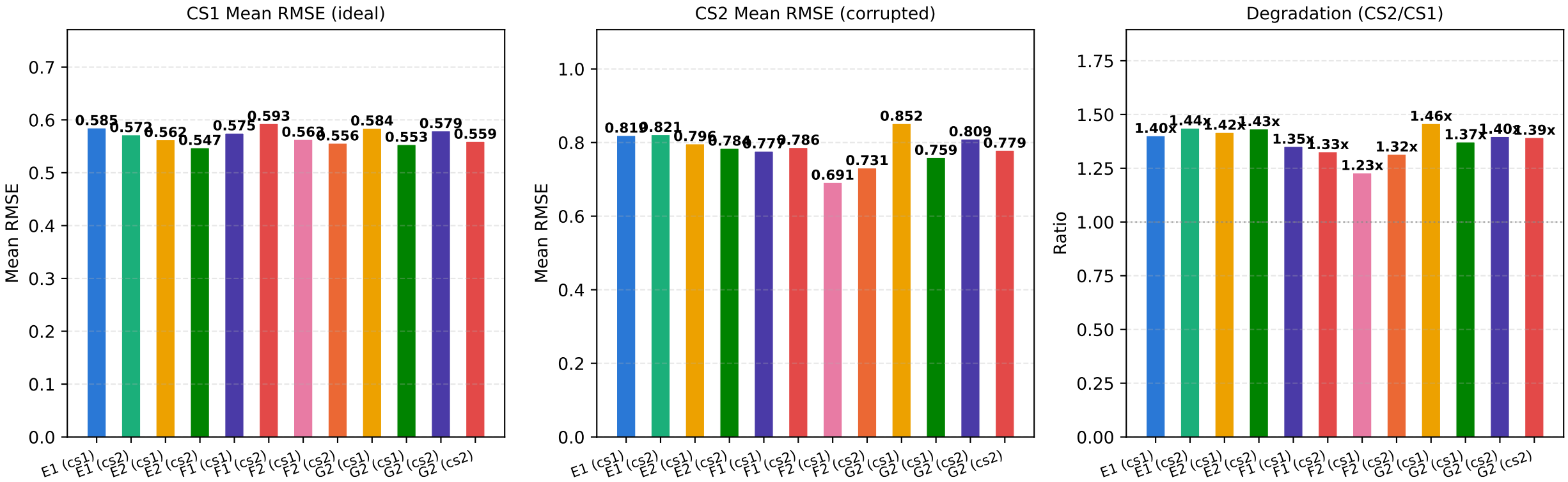
ID	Model	Trained On	X	Y	Z	Mean
E1_direct_unet_default_cs1	DirectUNet	cs1	0.4708	0.5651	0.7183	0.5847
E1_direct_unet_default_cs2	DirectUNet	cs2	0.4568	0.5487	0.7096	0.5717
E2_direct_unet_small_cs1	DirectUNet	cs1	0.4807	0.5308	0.6755	0.5623
E2_direct_unet_small_cs2	DirectUNet	cs2	0.4472	0.5183	0.6764	0.5473
F1_vanilla_cfm_default_cs1	VanillaCFM	cs1	0.4591	0.5279	0.7377	0.5749
F1_vanilla_cfm_default_cs2	VanillaCFM	cs2	0.4824	0.5458	0.7508	0.5930
F2_vanilla_cfm_small_cs1	VanillaCFM	cs1	0.4348	0.5186	0.7352	0.5628
F2_vanilla_cfm_small_cs2	VanillaCFM	cs2	0.4217	0.5114	0.7342	0.5558
G2_tweedie_small_cs1	Tweedie	cs1	0.4363	0.5686	0.7474	0.5841
G2_tweedie_small_cs1_k1	Tweedie	cs1	0.4437	0.5287	0.6873	0.5532
G2_tweedie_small_cs2	Tweedie	cs2	0.4205	0.5720	0.7449	0.5791
G2_tweedie_small_cs2_k1	Tweedie	cs2	0.4355	0.5344	0.7073	0.5591

--- CS2 ---

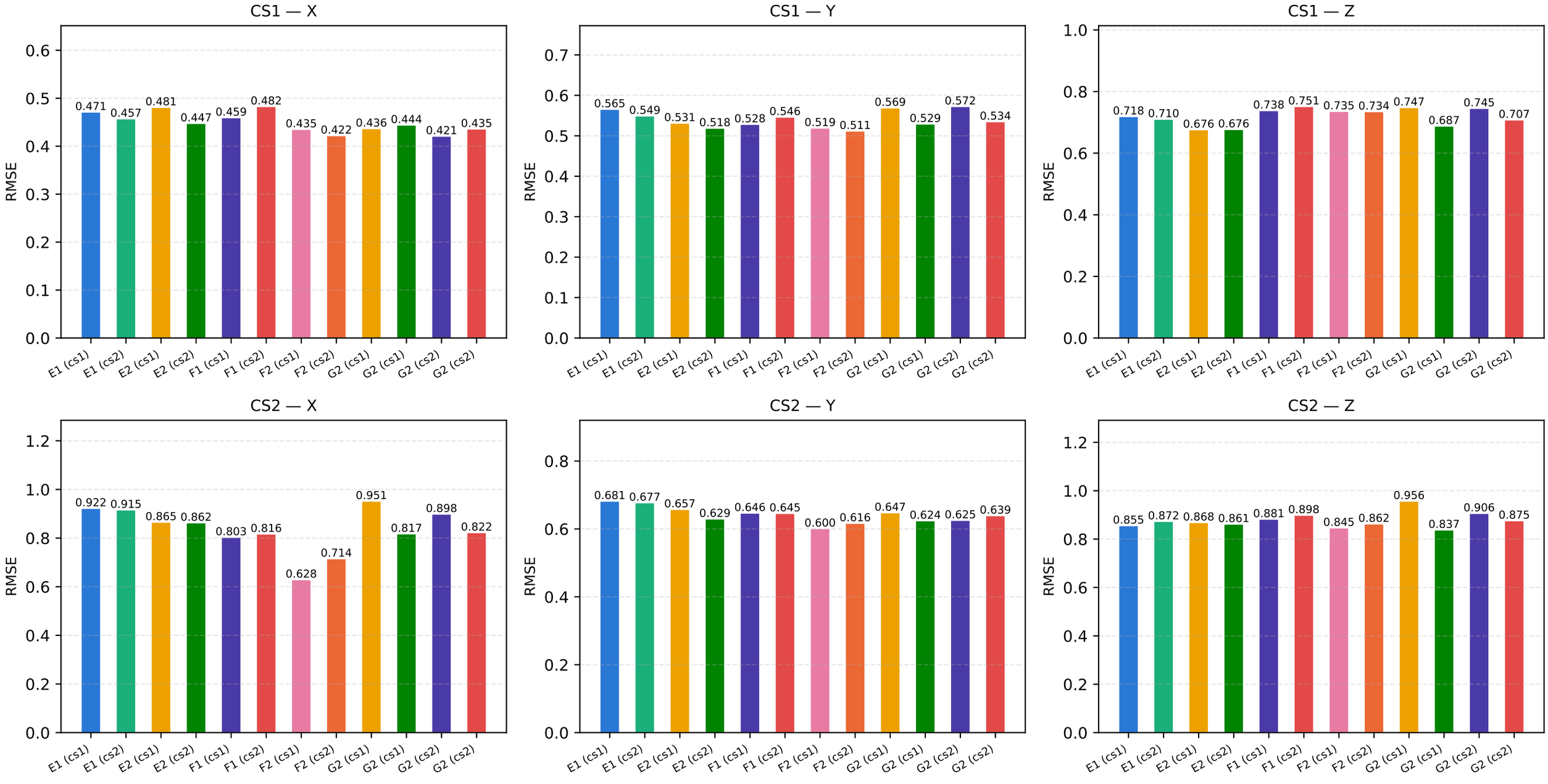
ID	Model	Trained On	X	Y	Z	Mean
E1_direct_unet_default_cs1	DirectUNet	cs1	0.9215	0.6814	0.8549	0.8193
E1_direct_unet_default_cs2	DirectUNet	cs2	0.9150	0.6766	0.8724	0.8214
E2_direct_unet_small_cs1	DirectUNet	cs1	0.8645	0.6570	0.8677	0.7964
E2_direct_unet_small_cs2	DirectUNet	cs2	0.8621	0.6289	0.8612	0.7841
F1_vanilla_cfm_default_cs1	VanillaCFM	cs1	0.8025	0.6461	0.8814	0.7767
F1_vanilla_cfm_default_cs2	VanillaCFM	cs2	0.8160	0.6452	0.8980	0.7864
F2_vanilla_cfm_small_cs1	VanillaCFM	cs1	0.6283	0.6004	0.8454	0.6914
F2_vanilla_cfm_small_cs2	VanillaCFM	cs2	0.7145	0.6160	0.8621	0.7309
G2_tweedie_small_cs1	Tweedie	cs1	0.9512	0.6471	0.9562	0.8515
G2_tweedie_small_cs1_k1	Tweedie	cs1	0.8166	0.6235	0.8371	0.7591
G2_tweedie_small_cs2	Tweedie	cs2	0.8979	0.6248	0.9055	0.8094
G2_tweedie_small_cs2_k1	Tweedie	cs2	0.8216	0.6386	0.8754	0.7785

Degradation (CS2 mean / CS1 mean) and training time:

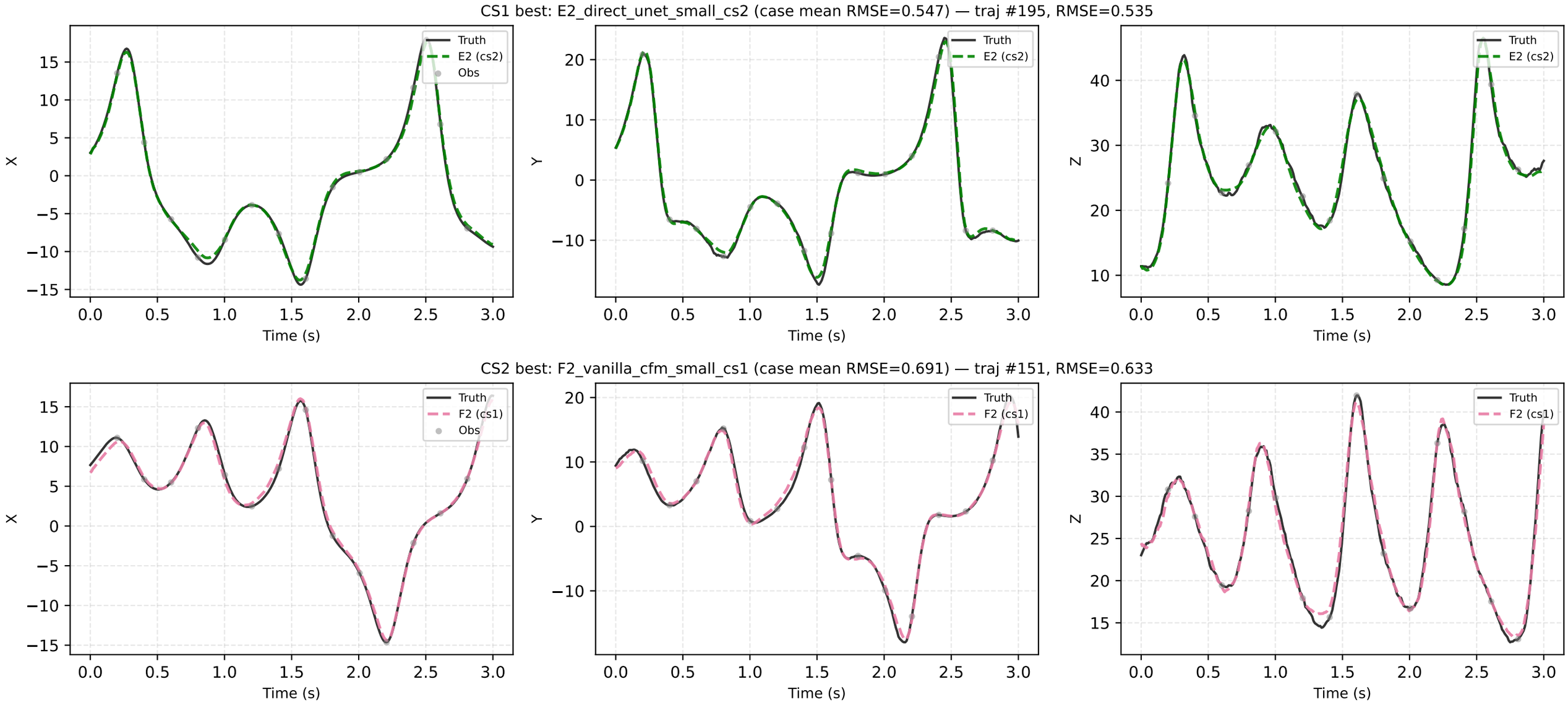
ID	Model	Trained On	CS1 mu	CS2 mu	Deg	Time(s)
E1_direct_unet_default_cs1	DirectUNet	cs1	0.5847	0.8193	1.40	x 8552
E1_direct_unet_default_cs2	DirectUNet	cs2	0.5717	0.8214	1.44	x 8547
E2_direct_unet_small_cs1	DirectUNet	cs1	0.5623	0.7964	1.42	x 8446
E2_direct_unet_small_cs2	DirectUNet	cs2	0.5473	0.7841	1.43	x 8451
F1_vanilla_cfm_default_cs1	VanillaCFM	cs1	0.5749	0.7767	1.35	x 11888
F1_vanilla_cfm_default_cs2	VanillaCFM	cs2	0.5930	0.7864	1.33	x 11872
F2_vanilla_cfm_small_cs1	VanillaCFM	cs1	0.5628	0.6914	1.23	x 11662
F2_vanilla_cfm_small_cs2	VanillaCFM	cs2	0.5558	0.7309	1.32	x 11671
G2_tweedie_small_cs1	Tweedie	cs1	0.5841	0.8515	1.46	x 19602
G2_tweedie_small_cs1_k1	Tweedie	cs1	0.5532	0.7591	1.37	x 10982


CS1 Mean RMSE (ideal)
Mean RMSE
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.585
0.573
0.562
0.547
0.575
0.593
0.563
0.556
0.584
0.553
0.579
0.559
E1 (cs1)
E1 (cs2)
E2 (cs1)
E2 (cs2)
F1 (cs1)
F1 (cs2)
F2 (cs1)
F2 (cs2)
G2 (cs1)
G2 (cs1)
G2 (cs2)
G2 (cs2)
CS2 Mean RMSE (corrupted)
Mean RMSE
0.0
0.2
0.4
0.6
0.8
1.0
0.819
0.821
0.796
0.784
0.777
0.786
0.691
0.731
0.852
0.759
0.809
0.779
E1 (cs1)
E1 (cs2)
E2 (cs1)
E2 (cs2)
F1 (cs1)
F1 (cs2)
F2 (cs1)
F2 (cs2)
G2 (cs1)
G2 (cs1)
G2 (cs2)
G2 (cs2)
Degradation (CS2/CS1)
Ratio
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
1.40x
1.44x
1.42x
1.43x
1.35x
1.33x
1.23x
1.32x
1.46x
1.37x
1.40x
1.39x
E1 (cs1)
E1 (cs2)
E2 (cs1)
E2 (cs2)
F1 (cs1)
F1 (cs2)
F2 (cs1)
F2 (cs2)
G2 (cs1)
G2 (cs1)
G2 (cs2)
G2 (cs2)

Per-Component RMSE Breakdown (CS1, CS2)



Example Reconstructions — Best Model per Case (median-RMSE trajectory)



Notes

```
Notes
======================================================================

Not yet trained (config exists, no results.json/loss log found):
  - G1_tweedie_default_cs1
  - G1_tweedie_default_cs2

Re-run this report after training to include them:
  python train.py --config-name experiment/G1_tweedie_default_cs1
  python train.py --config-name experiment/G1_tweedie_default_cs2
  python reports/generate_training_report.py
```